PHYSICS

SECTION-1

1) A quantity X is given by X = $\varepsilon_0 L \frac{\Delta V}{\Delta t}$ where ε^0 is the permittivity of free space, L is a length, ΔV is a potential difference and Δt is a time interval the dimensional formula for X is the same as that of :-

(1) Resistance
(2) Charge
(3) Voltage
(4) Current

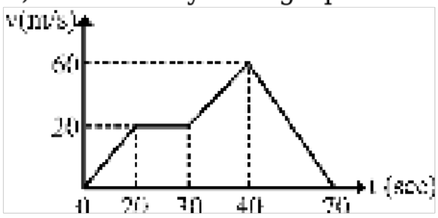
2) **Assertion :** Force cannot be added to pressure.
Reason : Physical quantities which have same dimensions can be added or subtracted together.

(1) Assertion and reason are true and the reason is the correct explanation of the assertion.
(2) Assertion and reason are true but reason is not the correct explanation of the assertion.
(3) Assertion is true but reason is false.
(4) Assertion and reason both are false.

3) A stone projected with a velocity u at an angle θ with the horizontal reaches maximum height H_1. When it is projected with velocity u at an angle $\left(\frac{\pi}{2}-\theta\right)$ with the horizontal, it reaches maximum height H_2. The relation between the horizontal range R of the projectile, H_1 and H_2 is :-

(1) R = $4\sqrt{H_1H_2}$
(2) R = $4(H_1 - H_2)$
(3) R = $4(H_1 + H_2)$
(4) R = $\frac{H_1^2}{H_2^2}$

4) The velocity-time graph of a body is shown in figure. The maximum acceleration in m/s^2 is:-



(1) 4
(2) 3

(3) 2

(4) 1

5) A body is thrown up in a lift with a velocity u relative to the lift and the time of flight is found to be t. The acceleration with which the lift is moving up is :

(1) $\frac{u-gt}{t}$

(2) $\frac{2u-gt}{t}$

(3) $\frac{u+gt}{t}$

(4) $\frac{2u+gt}{t}$

6) Which one of the following quantities has dimensions different from the remaining three

(1) Energy per unit volume

(2) Force per unit area

(3) Product of voltage and charge per unit volume

(4) Angular momentum per unit mass

7) The force F, on a sphere of radius r moving in a medium with velocity v is given by F = $\pi\eta rv$, then which of the following statements about the relation is correct (η = coefficient of viscosity):

(1) It is correct both dimensionally as well as numerically

(2) It is neither dimensionally correct nor numerically

(3) It is demensionally correct but not numerically

(4) It is numerically correct but not dimensionally

8) The displacement of a particle starting from rest (at t = 0) is given by $s = 6t^2 - t^3$. The time when the particle will attain zero acceleration is :

(1) 2s

(2) 8s

(3) 12s

(4) 16s

9) Two particles A and B are moving in vertical plane as shown in figure at t = 0. It is given that both particles collide mid air, then time after which A will collide to B will be –

(1) 4 s

(2) 5 s

(3) 2 s

(4) 8 s

10) The angular speed of a motor wheel is increased from 1200 rpm to 3120 rpm in 16 seconds, what is angular acceleration

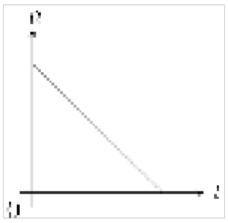
(1) $\frac{\pi}{2} \frac{\text{rad}}{\text{s}^2}$

(2) $\pi \frac{\text{rad}}{\text{s}^2}$

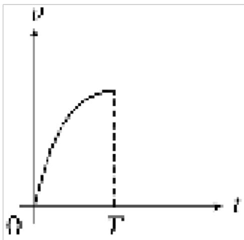
(3) $2\pi \frac{\text{rad}}{\text{s}^2}$

(4) $4\pi \frac{\text{rad}}{\text{s}^2}$

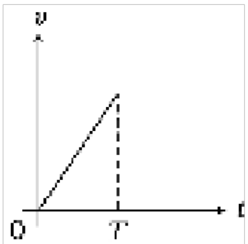
11) The acceleration-time (a – t) graph for a particle moving along a straight line starting from rest is shown in figure. Which of the following graph is the best representation of variation of its velocity (v) with time (t) ?



(1)



(2)



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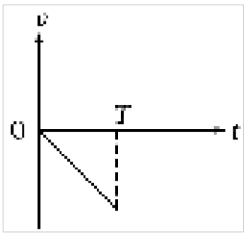

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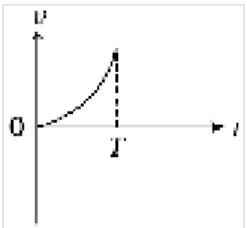


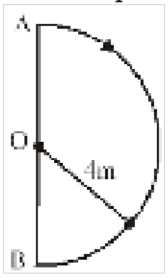
12) A wheel is subjected to uniform angular acceleration about its axis. Initially its angular velocity is zero. In the first 2 sec, it rotates through an angle θ_1 ; in the next 2 sec, it rotates through an

additional angle θ_2. The ratio of θ_2 / θ_1 is-

(1) 1
(2) 2
(3) 3
(4) 5

13) The work function of metal is w and λ is the wavelength of incident radiation there is no emission of photoelectrons when :-

(1) $\lambda > \frac{hc}{w}$
(2) $\lambda = \frac{hc}{w}$
(3) $\lambda < \frac{hc}{w}$
(4) $\lambda \leqslant \frac{hc}{w}$

14) In 2.0 second a particle goes from A to point B moving on a semi circle of radius 4m, the magnitude of average velocity is :- 

(1) 4 m/s
(2) 2 m/s
(3) 2π m/s
(4) zero

15) A projectile is projected with initial velocity $(6\hat{i} + 8\hat{j})$m/s. If g = 10 ms^{-2}, then horizontal range is:

(1) 4.8 metre
(2) 9.6 metre
(3) 19.2 metre
(4) 14.0 metre

16) The speed of a projectile at its maximum height is $\frac{\sqrt{3}}{2}$ times its initial speed. If the range of the projectile is P times the maximum height attained by it, P is equal to :-

(1) 4/3
(2) $4/\sqrt{3}$
(3) $4\sqrt{3}$

(4) 3/4

17) A projectile is given initial velocity of $\hat{i}+2\hat{j}$. The equation of the path is: (g = 10 m/s^2)

(1) $y = 2x - 5x^2$
(2) $y = x - 5x^2$
(3) $4y = 2x - 5x^2$
(4) $y = 2x - 25x^2$

18) If the momentum of an electron is changed by P, then the de Broglie wavelength associated with it changes by 0.8%. The initial momentum of electron will be :-

(1) 100 P
(2) 125 P
(3) 200 P
(4) 250 P

19)

Rain drops are falling with velocity $(2\hat{i}-3\hat{j})$ m/s what should be the velocity of a man so that rain drops hit him with speed 5 m/s

(1) $-\hat{i}$
(2) $5\hat{i}$
(3) $-2\hat{i}$
(4) $2\hat{i}$

20) In a vernier calliper, one main scale division is x cm and n divisions of the vernier scale coincide with (n–1) division of the main scale. The least count (in cm) of the calliper is :

(1) $\left(\frac{n-1}{n}\right)x$
(2) $\frac{nx}{(n-1)}$
(3) $\frac{x}{n}$
(4) $\frac{x}{(n-1)}$

21) A screw gauge gives the following reading when used to measure the diameter of a wire.
Main scale reading : 0 mm
Circular scale reading : 52 divisions
Given that 1mm on main scale corresponds to 100 divisions of the circular scale. The diameter of wire from the above data is :

(1) 0.026 cm
(2) 0.005 cm

(3) 0.52 cm

(4) 0.052 cm

22) An electron (mass m) with an initial velocity $v = v_0\hat{i}(v_0 > 0)$ is in an electric field $E = -E_0\hat{i}(E_0 = \text{constant} > 0)$ and has wavelength λ_0. Its de-Broglie wavelength at the time t is given by :

(1) $\dfrac{\lambda_0}{\left(1 + \frac{eE_0}{m}\frac{t}{v_0}\right)}$

(2) $\lambda_0\left(1 + \dfrac{eE_0t}{mv_0}\right)$

(3) λ_0

(4) $\lambda_0 t$

23) A nucleus of mass M at rest emits an α-particle of mass m. The de-Broglie wavelength of α-particle and recoil nucleus will be in the ratio?

(1) m : M

(2) (M + m) : m

(3) M : m

(4) 1 : 1

24) A projectile is projected with horizontal velocity (u) from height (h) then speed on striking the ground is:

(1) $\sqrt{u^2 + 2gh}$

(2) $\sqrt{2gh}$

(3) u

(4) u+2gh

25) A Body moves 6 m north. 8 m east and 10 m vertically upwards, what is its resultant displacement from initial position

(1) $10\sqrt{2}$ m

(2) 10 m

(3) $\dfrac{10}{\sqrt{2}}$ m

(4) 10 × 2 m

26) An experiment measures quantities x, y, z and then t is calculated from the data as $t = \dfrac{xy^2}{z^3}$. If percentage errors in x, y and z are respectively 1%, 3%, 2%, then percentage error in t is:

(1) 10 %

(2) 4 %

(3) 7 %

(4) 13 %

27) A cars starts from rest, moves with an acceleration a and then decelerates at a constant rate b for sometime to come to rest. If the total time taken is t, the maximum velocity of car is given by :-

(1) $\frac{abt}{(a+b)}$

(2) $\frac{a^2t}{(a+b)}$

(3) $\frac{at}{(a+b)}$

(4) $\frac{b^2t}{(a+b)}$

28) Threshold frequency for a metal is 10^{15} Hz. Light of λ = 4000 Å falls on its surface. Which of the following statements is correct ?

(1) No photoelectric emission takes place

(2) Photo-electrons come out with zero speed

(3) Photo-electrons come out with 10^3 m/sec speed

(4) Photo-electrons come out with 10^5 m/sec speed

29) The wavelength of light incident on a metal surface is reduced from 300 nm to 200 nm (both are less than threshold wavelength). What is the change in the stopping potential for photoelectrons emitted from the surface (Take $h = 6.6 \times 10^{-34}$ J-sec)

(1) 4 V

(2) 3 V

(3) 2 V

(4) 1 V

30) Preeti reached the metro station and found that the escalator was not working. She walked up the stationary escalator in time t_1. On other days, if she remains stationary on the moving escalator, then the escalator takes her up in time t_2. The time taken by her to walk up on the moving escalator will be

(1) $\frac{t_1t_2}{t_2-t_1}$

(2) $\frac{t_1t_2}{t_2+t_1}$

(3) $t_1 - t_2$

(4) $\frac{t_1+t_2}{2}$

31) Match the column :-

	Column-I		Column-II
(A)	$\frac{dx}{dt}$	(P)	$[M^0LT^0]$
(B)	$m\frac{d^2x}{dt^2}$	(Q)	$[M^1L^2T^{-2}]$
(C)	$\int Vdt$	(R)	$[M^0L^1T^{-1}]$
(D)	$\int a\,dt$	(S)	$[M^1L^1T^{-2}]$

x → Position of particle depending on time.
m → mass, V → velocity, a → acceleration

(1) (A) → (R), (B) → (S), (C) → (P), (D) → (Q)

(2) (A) → (R), (B) → (S), (C) → (P), (D) → (R)

(3) (A) → (R), (B) → (P), (C) → (Q), (D) → (S)

(4) (A) → (P), (B) → (R), (C) → (S), (D) → (Q)

32) A vernier callipers has 20 divisions on the vernier scale which coincide with 19 divisions on the main scale. The least count of the instrument is 0.2 mm. Then find out the value of main scale division :-

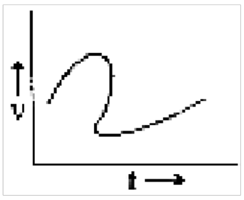
(1) 0.5 mm

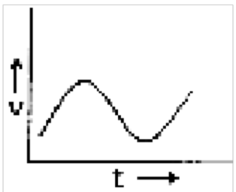
(2) 4 mm

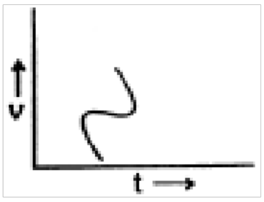
(3) 2 mm

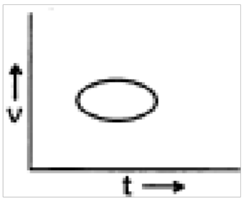
(4) $\frac{1}{4}$mm

33) Which of the following velocity - time graphs shows a realistic situation for a body in motion ?

(1) 

(2) 

(3) 

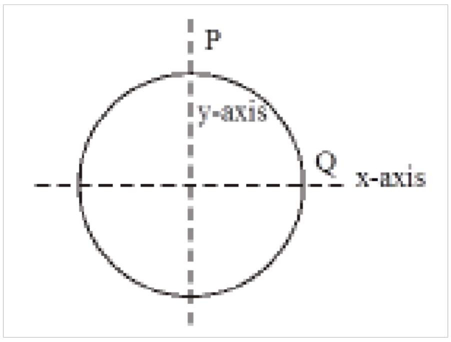
(4) 

34) When an α-particle of mass 'm' moving with velocity 'v' bombards on a heavy nucleus of charge 'Ze' its distance of closest approach from the nucleus depends on m as :

(1) $\frac{1}{m}$
(2) $\frac{1}{\sqrt{m}}$
(3) $\frac{1}{m^2}$
(4) m

35)

A particle moves in a circle of radius 4 cm clockwise at constant speed 2 cm/s. The acceleration of particle at instance half way between P & Q in cm/s^2 is :-



(1) $-4(\hat{i}+\hat{j})$
(2) $4(\hat{i}+\hat{j})$
(3) $-\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$
(4) $\left(\frac{\hat{i}-\hat{j}}{4}\right)$

SECTION-2

1) A body A is thrown up vertically from the ground with velocity v_0 and another body B is simultaneously dropped from a height H. They meet at a height $\frac{H}{2}$ if v^0 is equal to.

(1) $\sqrt{2gH}$
(2) $\sqrt{gH}$

(3) $\frac{1}{2}\sqrt{gH}$

(4) $\sqrt{\frac{2g}{H}}$

2) A person is standing on the top of a building of height 25 m. He want to throw a ball towards a boy who stands on top of another building of height 20 m at distance 15 m from first building. For which horizontal speed of projection, it is possible :

(1) 5 m/s

(2) 15 m/s

(3) 10 m/s

(4) 20 m/s

3)

If 13.6 eV energy is required to ionize the hydrogen atom, then the energy required to remove an electron from n = 2 is-

(1) 10.2 eV

(2) 0 eV

(3) 3.4 eV

(4) 6.8 eV

4) When a hydrogen atom is excited from ground state to first excited state then –
(a) its kinetic energy increase by 10.2 eV
(b) its kinetic energy decreases by 10.2 eV
(c) its potential energy increases by 20.4 eV
(d) its angular momentum increases by 1.05×10^{-34} J-s
Which of the statements are correct –

(1) b, c, d

(2) c, d

(3) a, c

(4) a, b, c, d

5) The speed of a boat is 5 km/hr in still water. If it crosses a river of width 1 km along the shortest possible path in 15 min., then velocity of the river is-

(1) 4 km/hr

(2) 3 km/hr

(3) 2 km/hr

(4) 1 km/hr

6) A particle is projected with a velocity v, so that its range on a horizontal plane is twice the greatest height attained. If g is acceleration due to gravity, then its range is :-

(1) $\frac{4v^2}{5g}$

(2) $\frac{4g}{5v^2}$

(3) $\frac{4v^3}{5g^2}$

(4) $\frac{4v}{5g^2}$

7) Energy levels A, B, C of a certain atom correspond to increasing values of energy i.e. $E_A < E_B < E_C$. If λ_1, λ_2, λ_3 are the wavelengths of radiation corresponding to the transitions C to B, B to A and C to A respectively, which of the following relation is correct ?

(1) $\lambda_3 = \lambda_1 + \lambda_2$

(2) $\lambda_3 = \lambda_1\lambda_2/(\lambda_1 + \lambda_2)$

(3) $\lambda_1 + \lambda_2 + \lambda_3 = 0$

(4) $\lambda_3^2 = \lambda_1^2 + \lambda_2^2$

8) An object is projected horizontally from a tower with initial velocity of 40 m/s. Find the time when final velocity will be making 30° with the horizontal :-

(1) 4 sec

(2) $\frac{2}{\sqrt{3}}$ sec

(3) $\sqrt{3}$ sec

(4) $\frac{4}{\sqrt{3}}$ sec

9) In the relation $P = \frac{\alpha}{\beta} e^{-\frac{\alpha Z}{k\theta}}$ P is pressure, Z is the distance, k is Boltzmann constant and θ is the temperature. The dimensional formula of β will be :-

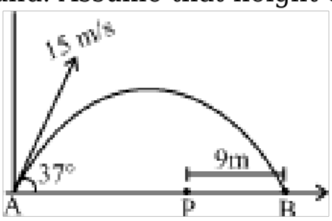
(1) $[M^0L^2T^0]$

(2) $[M^1L^2T^1]$

(3) $[M^1L^0T^{-1}]$

(4) $[M^1L^2T^{-1}]$

10) A ball is hit by a batsman at an angle of 37°. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile :-



(1) 3 m/s

(2) 5 m/s

(3) 9 m/s

(4) 12 m/s

11) A particle is dropped vertically from rest, from a height. The time taken by it to fall through successive distances of 1 km each will then be:-

(1) All equal, being equal to $\sqrt{2/g}$ second.

(2) In the ratio of the square roots of the integers $1 : \sqrt{2} : \sqrt{3}$

(3) In the ratio of the difference in the square roots of the integers, i.e., $\sqrt{1} : \left(\sqrt{2} - \sqrt{1}\right) : \left(\sqrt{3} - \sqrt{2}\right) : \left(\sqrt{4} - \sqrt{3}\right)$........

(4) In the ratio of the reciprocals of the square roots of the integers, i.e., $\frac{1}{\sqrt{1}} : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$......

12) The retardation of a moving particle, if the relation between time and position is $t = x^2 + x$ will be-

(1) $2(x + 1)^{-3}$

(2) $2(2x + 1)^{-3}$

(3) $1/2\,(x + 1)^{-3}$

(4) $1/2\,[2x + 1]^{-3}$

13) A body starts from rest and travels a distance S with uniform acceleration, then moves uniformly a distance 2S and finally comes to rest after moving further 3S under uniform retardation. The ratio of the average velocity to maximum velocity is :-

(1) 2/5

(2) 3/5

(3) 4/7

(4) 5/7

14) Trajectory of particle in a projectile is given as $y = x - \frac{x^2}{80}$. Here, x and y are in metre. For this projectile motion match the following and select proper option ($g = 10\ m/s^2$)

	Column-I		Column-II
A	Angle of projection	p	20 m
B	Angle of velocity with horizontal after 4s	q	80 m
C	Maximum height	r	45°
D	Horizontal range	s	$\tan^{-1}\left(\frac{1}{2}\right)$

(1) A → s, B → s, C → q, D → p

(2) A → r, B → r, C → q, D → p

(3) A → r, B → r, C → p, D → q

(4) A → s, B → r, C → p, D → q

15) Two balls A and B are moving with velocity $(\hat{i}+\hat{j})$ m/s and $(4\hat{i}-3\hat{j})$ m/s respectively. Then the magnitude of relative velocity of ball B w.r.t. ball A is

(1) 3 m/s

(2) 4 m/s

(3) $\sqrt{41}$ m/s

(4) 5 m/s

CHEMISTRY

SECTION-1

1) 3A + 4B → C, if 1 mol A and 4 mol B are reacted, find mass of C formed.
(Molecular weight of C = 24)

(1) 24 g

(2) 10 g

(3) 8 g

(4) 12 g

2) The number of atoms in 4.25 g of NH_3 is approximately

(1) 4×10^{23}

(2) 2×10^{23}

(3) 1×10^{23}

(4) 6×10^{23}

3) The simplest formula of a compound containing 50% of the element X (At. mass = 10) and 50% of the element Y (at mass = 20) is :-

(1) XY

(2) X_2Y

(3) XY_2

(4) X_2Y_3

4) The molarity of a glucose solution containing
36 g of glucose per 400 mL of the solution is

(1) 1.0

(2) 0.5

(3) 2.0

(4) 0.05

5) What mass of 95% pure $CaCO_3$ will be required to neutralise 50 mL of 0.5 M HCl solution according to the following reaction?
$CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$
[Calculate upto second place of decimal point]

(1) 1.32 g

(2) 3.65 g

(3) 9.50 g

(4) 1.25 g

6) Which of the following sets of compounds illustrates the law of multiple proportion ?

(1) Na_2O, K_2O, Li_2O

(2) NaCl, KCl, HCl

(3) CO, CO_2, CH_4

(4) None

7) The e/m ratio is maximum for :

(1) D^+

(2) He^+

(3) H^+

(4) He^{2+}

8) The number of electrons in nucleus of ${}^{14}_{7}N$ will be :

(1) 14

(2) 7

(3) 10

(4) Zero

9) For which of the following transitions would a hydrogen atom absorb a photon with longest wavelength?

(1) n = 1 to n = 2

(2) n = 3 to n = 2

(3) n = 5 to n = 6

(4) n = 7 to n = 6

10) Which of the following series of transitions in the spectrum of hydrogen atom falls in visible region?

(1) Brackett series
(2) Lyman series
(3) Balmer series
(4) Paschen series

11) Minimum de-Broglie wavelength is associated with:-

(1) Electron
(2) Proton
(3) CO_2 molecule
(4) SO_2 molecule

12) A compound of vanadium possesses a magnetic moment of 1.73 BM. The oxidation state of vanadium in this compounds is :-

(1) +4
(2) +3
(3) +2
(4) +1

13) Which of the following is a liquid non metal ?

(1) Hg
(2) Ga
(3) Bromine
(4) Cs

14) The element having electronic configuration $[Xe]4f^{14}6s^2$ belongs to :

(1) s-block
(2) p-block
(3) d-block
(4) f-block

15) Select the amphoteric substance in the following

(1) SO_3
(2) NaOH
(3) CO_2
(4) $Al(OH)_3$

16) Reason of lanthanoid contraction is :-

(1) Negligible screening effect of 'f' orbitals
(2) Increasing nuclear charge

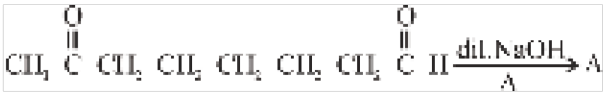
(3) Decreasing nuclear charge
(4) Increasing screening effect

17) The most polar bond is present in :-

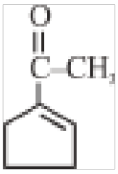
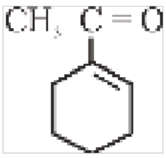
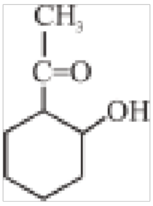
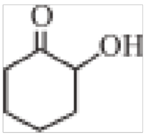
(1) CH_4
(2) NH_3
(3) H_2O
(4) HF

18) Correct order of atomic size is:-

(1) $Li^+ < F^- < O^{2-} < Na^+$
(2) $Li^+ < O^{2-} < F^- < Na^+$
(3) $Li^+ < Na^+ < F^- < O^{2-}$
(4) $Li^+ < Na^+ < O^{2-} < F^-$

19) 

Product A is :-

(1) 
(2) 
(3) 
(4) 

20)

Which of the following will not give aldols condensation?

(1) Acetaldehyde
(2) Propenaldehyde
(3) Benzaldehyde
(4) Acetone

21) Tollen’s reagent and Fehling solution are used to distinguish between.

(1) Acid and Alcohol
(2) Alkane and Alcohol
(3) n-alkane and branched alkane
(4) Ketones and Aldehydes

22) The correct order of boiling point of following compounds is-
(I) $CH_3CH_2CH_2CH_2OH$
(II) $CH_3CH_2CH_2CHO$
(III) $CH_3CH_2CH_2COOH$

(1) I > II > III
(2) III > II > I
(3) III > I > II
(4) I > III > II

23) Which among the following undergoes haloform test but cannot give Fehling’s test ?

(1) Ethanal
(2) Propanone
(3) Ethanoic acid
(4) Butan-1-ol

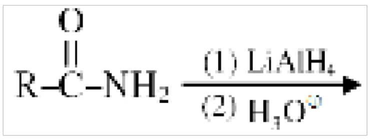
24) In Etard's reaction toluene is oxidised to benzaldehyde using-

(1) H_2O_2
(2) Cl_2
(3) CrO_2Cl_2
(4) $KMnO_4$

25) 

What can be x ?

(1) Red P + HI
(2) Zn–Hg + HCl
(3) $H_2N–NH_2/OH^-$
(4) All of the above

26) 

P (Major)
P should be :-

(1) R-CH_2-OH

(2) R-CH_2-NH_2

(3) 

(4) RCOOH

27) $CH_3CH_2COOH \xrightarrow[\text{Red P}]{Br_2} X \xrightarrow[\Delta]{\text{Alc. KOH}} Y$

Y in the above reaction is :-

(1) CH_3COCl

(2) CH_3CH_2CHO

(3) CH_2=CHCOOH

(4) Cl-CH_2-CH_2-COOH

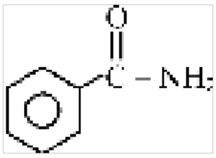
28) Primary and secondary amines can be distinguished by :-

(1) Carbylamine reaction

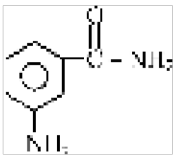
(2) Hinsberg reagent

(3) Both (1) & (2)

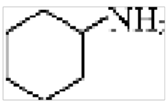
(4) Lucas test

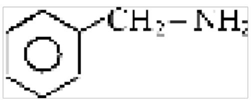
29)  $\xrightarrow[\Delta]{\text{(i) NaOH/Br}_2}$ (A)

Product A is :-

(1) 

(2) 

(3) 

(4) 

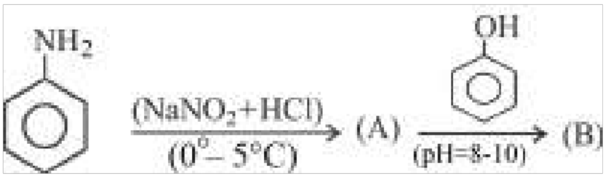
30) Correct statement is:-

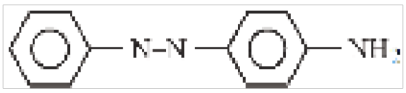
(1) $CH_3OCH_2CH_2CH_3$ and $CH_3CH_2OCH_2CH_3$ are metamers

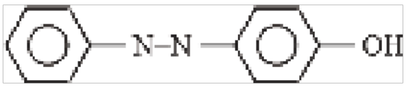
(2) Reactivity of alcohols for Lucas reagent follows the order:
3° alcohol > 2° alcohol > 1° alcohol

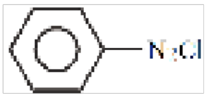
(3) Carbene is electron deficient species

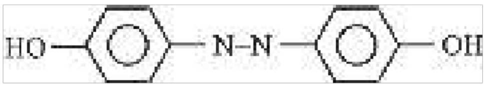
(4) All of the above

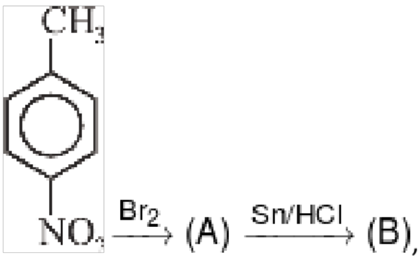
31) 
What will be (B) ?

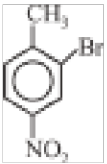
(1) 

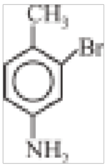
(2) 

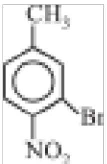
(3) 

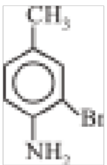
(4) 

32) , Find (B) :-

(1) 

(2) 

(3) 

(4) 

33) The reagent with which both acetaldehyde and acetophenone react easily are

(1) Tollen's reagent

(2) 2, 4-Dinitrophenyl hydrazine

(3) Fehling's solution

(4) Schiff's reagent

34) **Assertion (A):-** Carboxylic acids are more acidic than phenols.
Reason (R):- The carboxylate ion is highly unstable.

(1) Both (A) and (R) are correct; (R) is the correct explanation of (A)
(2) Both (A) and (R) are correct; (R) is not the correct explanation of (A)
(3) (A) is correct; (R) is incorrect
(4) (R) is correct; (A) is incorrect

35) Consider the following reaction,

The product A is :

(1) C_6H_5CHO
(2) C_6H_5OH
(3) $C_6H_5COCH_3$
(4) C_6H_5Cl

SECTION-2

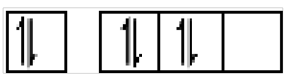
1) At NTP 5.6 L of a gas weighs 60g. The vapour density of the gas is :

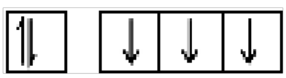
(1) 60
(2) 120
(3) 30
(4) 240

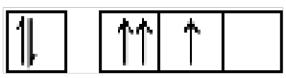
2) A gaseous mixture contains oxygen and ozone in the ratio of 1 : 4 by mass. The ratio of their number of atoms is :-

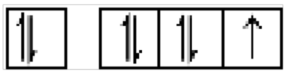
(1) 1 : 4
(2) 1 : 8
(3) 3 : 8
(4) 2 : 3

3) The orbital diagram in which both the Pauli's exclusion principle and Hund's rule are violated is:-

(1) 

(2) 

(3) 

(4) 

4) Two nuclides X and Y are isotonic with each other with mass numbers 39 and 40 respectively. If the atomic number of X is 19, that of Y is :-

(1) 20
(2) 19
(3) 21
(4) 18

5) Wave number of radiations having frequency of 4×10^4 Hz will be :

(1) 1.33×10^{-6} cm^{-1}
(2) 1.33×10^{-7} cm^{-1}
(3) 9×10^{-11} cm^{-1}
(4) 4×10^{-5} cm^{-1}

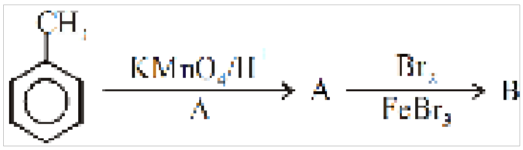
6)

An ion M^{2+} has electronic configuration [Ar] $3d^{10}$ $4s^2$ element M belongs to

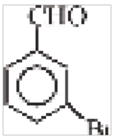
(1) s-Block
(2) p-Block
(3) d-Block
(4) f-Block

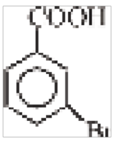
7) Correct increasing order of acidic character is :-

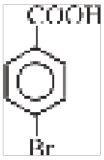
(1) $Al_2O_3 < MgO < Na_2O < Cs_2O$
(2) $Cs_2O < Na_2O < MgO < Al_2O_3$
(3) $Na_2O < MgO < Al_2O_3 < Cs_2O$
(4) None of these

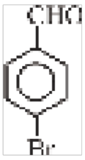
8) 

Identify product B :-

(1) 

(2) 

(3) 

(4) 

9) Which of these do not contain –COOH group

(1) Aspirin
(2) Benzoic acid
(3) Picric acid
(4) Salicylic acid

10) **Assertion :** Formaldehyde is a planar molecule.
Reason : It contains sp^2 hybridised carbon atom.

(1) Assertion and reason both are correct and reason is correct explanation of assertion.
(2) Assertion and reason both are wrong statements.
(3) Assertion is correct statement but reason is wrong statement.
(4) Assertion is wrong statement but reason is correct statement.

11) In which of the following reaction carbon-carbon bond is not form during the formation of product.

(1) Reimer-Tiemann Reaction
(2) Wurtz Reaction
(3) Cannizaro Reaction
(4) Friedel Craft Reaction

12)

The reaction involving the treatment of benzene diazonium chloride with CuCl is treated as

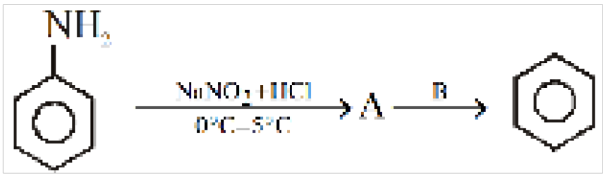
(1) Sandmeyer's reaction
(2) Wurtz's reaction
(3) Swarts reaction
(4) Finkelstein reaction

13) 1°, 2° and 3° amines are differentiated by :-

(1) Hinsberg's test
(2) Lucas test
(3) Lieberman nitroso test
(4) Iodoform test

14) **Statement-I :** Aromatic amines are generally less basic than ammonia and aliphatic amine.
Statement-II : Generally in substituted aniline, it is observed that electron releasing groups increase basic strength whereas electron withdrawing groups decrease it.

(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-I is incorrect but Statement-II is correct.

15) 

Reagent B can be :-

(1) H_3PO_2
(2) $C_2H_5N_2Cl$
(3) CH_3OH
(4) $SnCl_2$ + HCl

BOTANY

SECTION-1

1) Match the columns :-

	Column-I		Column-II
a.	Metacentric	i.	Terminal centromere
b.	Submetacentric	ii.	Centromere is situated close to its end
c.	Acronematic	iii.	Middle centromere
d.	Telocentric	iv.	Centromere is slightly away from middle

(1) a–i, b–iv, c–ii, d–iii
(2) a–ii, b–iv, c–i, d–iii
(3) a–iii, b–iv, c–ii, d–i
(4) a–iv, b–iii, c–i, d–ii

2) Condensation of chromosomes is completed at which phase of cell division?

(1) Prophase
(2) Metaphase
(3) Anaphase
(4) Telophase

3) If at the end of meiosis, the four daughter cells have 4 chromosomes, how many chromosomes were there in the mother cell ?

(1) 8
(2) 16
(3) 2
(4) 3

4) Which of the following statement is wrong ?

(1) Crossing over is an enzyme mediated process
(2) Dissolution of synaptonemal complex occure in diplotene
(3) Amount of DNA increases in s-phase
(4) Division of centromere occurs during anaphase-I

5) Diakinesis marked by

(1) Chiasmata clearly visible
(2) Crossing over
(3) Terminalisation of chiasmata
(4) Formation of synaptonemal complexes.

6) If the number of chromosomes in S-phase is 24 then what will be the number of chromosome present in anaphase of mitosis :-

(1) 24
(2) 36
(3) 48
(4) 96

7) Once nerve cells become mature, they do not usually undergo cell division. Based on your knowledge of the cell cycle, you would predict that mature nerve cell is present in which phase of cell cycle

(1) G_2-phase
(2) M-phase
(3) G_1-phase
(4) G_0-phase

8) The key features of metaphase are

(1) Spindle fibres attach to kinetochores of chromosomes

(2) Chromosomes are moved to spindle equator and get aligned along metaphase plate

(3) Splitting of centromere

(4) Both 1 and 2

9) In oocytes of some vertebrates which stage can last for months or years:

(1) Diakinesis

(2) Pachytene

(3) Diplotene

(4) None of these

10) If root cell of a plant contain 20 chromosome, then how many chromatids, the cell will have at anaphase-I of meiosis-I :-

(1) 10

(2) 20

(3) 40

(4) 80

11) Which stage of meiosis - I, the four chromatids of each bivalent chromosomes becomes distinct and clearly appears as tetrads

(1) Zygotene

(2) Pachytene

(3) Diplotene

(4) Diakinesis

12) In plant and animal cells the division of cytoplasm is respectively :-

(1) Centripetal and centripetal

(2) Centrifugal and centripetal

(3) Centripetal and centrifugal

(4) Centrifugal and centrifugal

13) Biotic potential is

(1) Intrinsic rate of natural increase under environmental limited condition

(2) Intrinsic rate of natural increase under environmental unlimited condition

(3) Extrinsic rate of natural increase under environmental limited conditions

(4) Extrinsic rate of natural increase under environmental unlimited conditions

14) Which of the following association shows mutualism?

(1) Fig and wasp

(2) Barnacles on whale

(3) Roundworms in human intestine

(4) Orchids on mango tree

15) The declining phase of a population occurs when

(1) Natality > Mortality

(2) Mortality > Natality

(3) Mortality = Natality

(4) Natality = Mortality = 0

16) The proportion of young individuals is highest in

(1) Expanding population

(2) Declining population

(3) Stable population

(4) Both stable and declining population

17) Predators help in :

(1) Maintaining species diversity

(2) Reduce the competition between prey species

(3) Maintaining ecological balance

(4) All

18) A biologist studied the population of rats in a barn. He found that the average natality was 250, average mortality 240, immigration 20 and emigration 30. The net increase in population is

(1) 0

(2) 15

(3) 5

(4) 10

19) Which of the following is **WRONG** ?

(1) Lichen, an association of fungus and alga, is an example of mutualism

(2) Epiphytes, using other plants for support only and not for water of food, show commensalism

(3) Parasite have well developed digestive system

(4) Mutualism included under positive interactions

20) The process of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration is referred as

(1) Immigration

(2) Association

(3) Emigration

(4) Competition

21) Which of the following is not an attribute of a population?

(1) Natality
(2) Mortality
(3) Species interaction
(4) Sex ratio

22) If '8' Drosophilla in a laboratory population of '80' died during a week, the death rate in the population is ___ individuals per Drosophilla per week.

(1) 1
(2) 0
(3) 0.1
(4) 10

23) Asymptote in a Logistic growth curve is obtained when

(1) The value of 'r' approaches zero
(2) $K < N$
(3) $K > N$
(4) $K = N$

24) Which one is correct for population growth rate of logistic growth population?

(1) $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$
(2) $\frac{dN}{dt} = rN$
(3) $N_t = N_0 e^{rt}$
(4) $\frac{dN}{dt} = \left(\frac{K-N}{K}\right)$

25) Gause's principle of competitive exclusion states that

(1) no two species can occupy the same niche indefinitely for the same limiting resources
(2) larger organisms exclude smaller ones through competition
(3) more abundant species will exclude the less abundant species through competition
(4) competition for the same resources exclude species having different food preferences

26) Consider the following statements.
Statement-I : All living organisms are composed of cells and product of cells.
Statement-II : Omnis cellula-e-cellula.

(1) Both I and II are correct
(2) Both I and II are incorrect

(3) I is correct and II is incorrect
(4) I is incorrect while II is correct

27) In which of the following processes/functions, there is no role of mesosome?

(1) DNA replication
(2) Distribution of daughter cells
(3) Cell wall formation
(4) Movement of cilia and flagella

28) Proteins, present in plasma membrane are classified as integral and peripheral on the basis of :-

(1) Density and size
(2) ease of extraction from membrane
(3) Structure
(4) Quantity

29) Lipids are arranged within cell membrane with

(1) polar heads towards inner side and the hydrophobic tails towards outerside
(2) both heads and tails towards outerside.
(3) polar heads towards outerside and tails towards inside.
(4) both heads and tails towards inner side.

30) Arrangement of microtubules in axoneme of cilia :-

(1) 9 double + 0
(2) 9 singlet + 9 doublet
(3) 9 singlet + 2 doublet
(4) 9 doublet + 2 singlet

31) ATP synthase are not found in :

(1) Mitochondria
(2) Chloroplast
(3) (1) and (2) Both
(4) Nuclear membrane

32) **Statement-I :** Plastids are found in all plant cells and in Euglenoides.
Statement-II : Majority of the chloroplasts of the green plants are found in the mesophyll cells of the leaves.

(1) Statement-I and statement-II both are true
(2) Statement-I is true and statement-II is false
(3) Statement-I is false and statement-II is true
(4) Statement-I and statement-II both are false

33) Which of the following is true for centriole ?

(1) Two centrioles of a centrosome are parallel to each other

(2) They form basal body of flagella in prokaryotic cell

(3) Each of the peripheral fibril in it is a triplet

(4) They are responsible for formation of spindle apparatus in plant cells

34)

An elaborate network of filamentous _____ structure present in the cytoplasm is collectively referred to as the ....

(1) Cytoskeleton, Proteinaceous

(2) Proteinaceous, Cytoskeleton

(3) Plasma membrane, Proteinaceous

(4) Endoskeleton, Proteinaceous

35) Given below are two statements :
Statement-I : At number of places the nuclear envelop is interrupted by minute pores.
Statement-II : Material of the nucleus stained by the acidic dye was given the name chromatin by Flamming. In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement I and statement II are correct

(2) Both statement I & statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

SECTION-2

1) (i) Chromosomes start pairing together
(ii) Chromosome synapsis is accompanied by the formation of synaptonemal complex.
(iii) Bivalents are very clearly visible as tetrads.
(iv) Bivalents are formed
From the above statements, which statement/(s) characterize the second phase of prophase-I

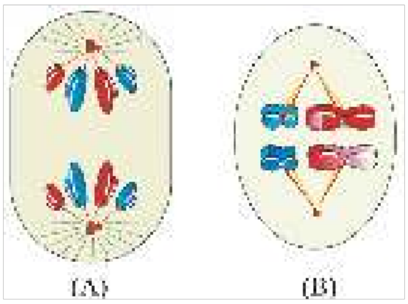
(1) All except iii

(2) Only i & iv

(3) Only i & ii

(4) All

2) Identify the diagram A and B and select the right option regarding them :-



	A	B
(1)	Anaphase–I	Metaphase–I
(2)	Anaphase	Metaphase–II
(3)	Anaphase	Metaphase–I
(4)	Anaphase–I	Metaphase–II

(1) 1

(2) 2

(3) 3

(4) 4

3) Which of the following is true for telophase?

(1) Reformation of nuclear envelope, Golgi, nucleolus and ER occurs

(2) Chromosomes cluster at opposite spindle poles

(3) Chromosomes lose their identity as discrete units

(4) More than one option is correct

4) Read the statements below :-
a. In the 24 hour average duration of cell cycle of a human cell, cell divisions lasts for only about an hour.
b. Interphase lasts more than 95% of the duration of cell cycle.
c. DNA content gets half at S phase of interphase.
d. G_2 phase is also called post mitotic phase.
Select the correct option :

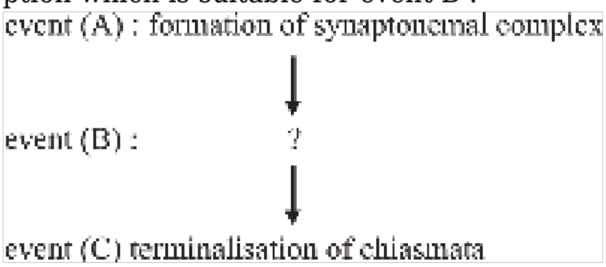
(1) a and b

(2) b, a and c

(3) a, b and d

(4) c, a and d

5) The three events (A, B and C) of cell cycle are given below in systematic way, select the correct option which is suitable for event B :-



Options :-

(1) Division of centromere

(2) Movement of chromosomes towards cell equator

(3) Crossing over between non homologous chromosomes

(4) Activity of recombinase enzyme

6) When does the growth rate of a population following the logistic model equal zero ? The logistic model is given as dN/dt = rN(1–N/K) :-

(1) when N/K is exactly one.

(2) when N nears the carrying capacity of the habitat.

(3) when N/K equals zero.

(4) when death rate is greater than birth rate.

7) **Assertion (A) :** Under normal conditions, births and deaths are the most important factors influencing population density.
Reason (R) : Immigration and emigration assuming importance only under special conditions.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) Both A and R are false statements

8) How many statement are correct ?
(a) Female mosquito is considered as true parasite.
(b) Galapagos island was inhabited by abingdon tortoise and those were eliminated by goats.
(c) *Calotropis* produces cardiac glycoside hence are poisonous plant.
(d) Clown fish are benefitted when present around sea anemone.

(1) Two

(2) Three

(3) Four

(4) One

9) Which of the following are correct matching pairs ?

(a)	Competition exclusion	(i)	Pisaster
(b)	Resource partition	(ii)	Ectoparasite
(c)	Predation	(iii)	Barnacle
(d)	Copepods	(iv)	Warblers

(1) a–iv, b–iii, c–ii, d–i

(2) a–iii, b–iv, c–i, d–ii

(3) a–ii, b–i, c–iii, d–iv

(4) a–i, b–ii, c–iv, d–iii

10) If all the starfish were removed from an area of the American pacific coast then:

(1) more than 10 species of invertebrates become extincted with a year.

(2) No any species extincted

(3) 10 species of invertebrates increases within a year

(4) All of the above

11) Given below are two statements :
Statement-I : Ribosomes are membrane-less organelles.
Statement-II : Cytoplasm is the main area of cellular activities.
In the light of the above statements choose the correct answer from options given below.

(1) Both Statement-I and Statement-II are correct.

(2) Both Statement-I and Statement-II are incorrect.

(3) Statement-I is correct and Statement-II is incorrect.

(4) Statement-I is incorrect but Statement-II is correct.

12) **Assertion :** Middle lamella holds or glues the different neighbouring cells together.
Reason : The middle lamella is mainly made up of Mg pectate.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

13) True statements for endomembrane system is:
A. Function of ribosomes are coordinated with ER so ribosomes are part of endomembrane system
B. Mitochondria and chloroplast are not the part of endomembrane system
C. Different membrane bound cell organelles which have coordinated functions are included in endomembrane system
D. Vacuole is not a part of endomembrane system

(1) A, B, C and D

(2) Only B and D

(3) Only B and C

(4) Only A, B and C

14) i. Non membrane bound organelles
ii. found in all cells
iii. found in cytoplasm, chloroplast, mitochondria and on RER of cell
These characters are belongs to which cell organelles.

(1) Centriole

(2) Vacuole

(3) Lysosome

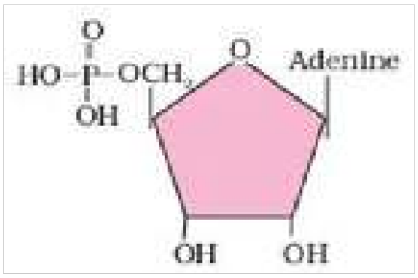
(4) Ribosome

15) **Assertion :** The content of nucleolus is continuous with the rest of the nucleoplasm.
Reason : Nucleolus is not a membrane bound structure.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

ZOOLOGY

SECTION-1

1) Identify the structure given below. 

(1) Nucleotide - Adenosine
(2) Nucleoside - Adenosine
(3) Nucleotide - Adenylic acid
(4) Nucleoside - Adenylic acid

2) Find out the wrongly matched pair :

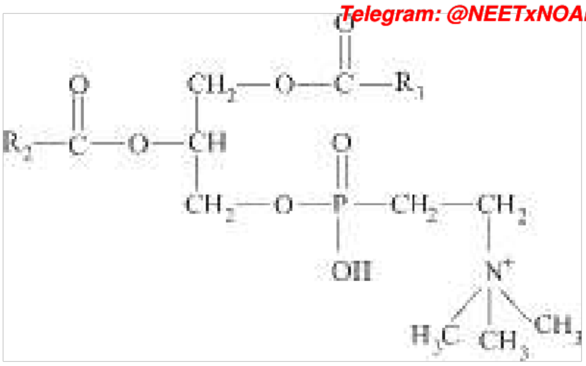
(1) Primary metabolite - ribose
(2) Protein – insulin
(3) Secondary metabolite – anthocyanins
(4) Cellulose – heteropolymer

3)

Match the protein given in Column I with its function given in Column II and choose the right option.

Column - I (Proteins)		**Column - II (Functions)**	
A.	Collagen	I.	Glucose transport
B.	Trypsin	II.	Hormone
C.	Insulin	III.	Intercellular ground substance
D.	GLUT - 4	IV.	Enzyme

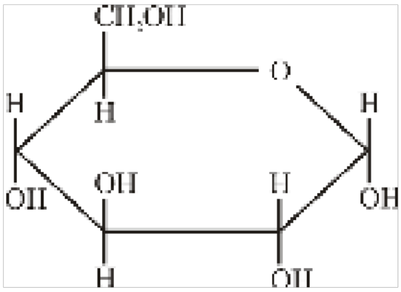
(1) A – III, B – IV, C – II, D – I
(2) A – IV, B – I, C – II, D – III
(3) A – II, B – IV, C – I, D – III
(4) A – III, B – IV, C – I, D – II

4) Following structure is related to which compound? 

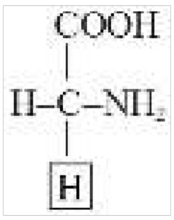
(1) Simple lipid
(2) Glycolipid
(3) Lecithin
(4) Chromolipid

5) Primary metabolites:

(1) Include glucose and fructose
(2) Present in all living tissues
(3) Plays known roles in all physiological process
(4) All of the above

6) 

(1) It is an example of hexose sugar
(2) Structure is α–D–Glucose
(3) Reduces fehling and Benedict solution
(4) All of the above

7) Identify the following: 

(1) Glycine
(2) Valine
(3) Leucine
(4) Methionine

8) Assertion A: Amino acids can exist in a zwitterionic form due to the ionizable nature of their

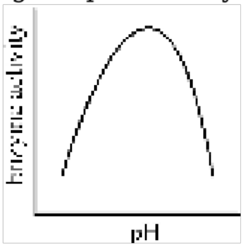
amino and carboxyl groups.
Reason R: In solutions of different pH, the structure of amino acids does not change.

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

9) Given below are two statements:
Statement I: Paper made from plant pulp and cotton fibre is starch and consists of complex polysaccharides.
Statement II: Chitin, found in the exoskeletons of arthropods, is an example of a complex polysaccharide that is mostly a homopolymer.
In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is false but Statement II is true.
(2) Both Statement I and Statement II are true.
(3) Both Statement I and Statement II are false.
(4) Statement I is true but Statement II is false.

10) Represented below is the effect of change in pH on enzyme activity. Which one of the following conclusions can be made by this diagram ?

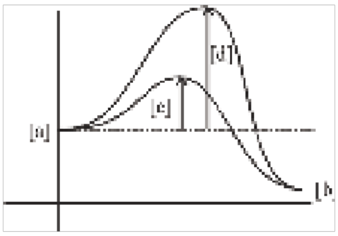


(1) Enzyme shows its highest activity at a particular pH, both below and above this pH, activity declines
(2) With increasing pH, activity increases upto a limit then it become constant
(3) At a particular pH, activity is minimum, both below and above this pH activity increases
(4) Enzyme activity is affected by temperature not by pH so it is completely a hypothetical graph

11) Identify the incorrect match :-

	Column-I	Column-II
(1)	Hexokinase	Transferase
(2)	Aldolase	Oxidoreductase
(3)	Transfer of 'H'	Oxidoreductase
(4)	Synthases	Ligases

(1) 1
(2) 2
(3) 3
(4) 4

12) In the following figure identify a, b, c & d :- 

(1) a-Product, b-substrate, c-activation energy with enzyme, d-activation energy without enzyme.

(2) a-Substrate, b-product, c-activation energy with enzyme, d-activation energy without enzyme.

(3) a-Substrate, b-product, c-activation energy without enzyme, d-Activation energy with enzyme.

(4) a-Substrate, b-product, c-activation energy with catalyst, d-activation energy with enzyme.

13)  $\longrightarrow$ X – Y + C = C

Above give reaction is catalysed by enzymes of class :

(1) Isomerase

(2) Lyases

(3) Hydolases

(4) Transferase

14) When the binding of the chemical shuts off enzyme activity, the process and the chemical are respectively called as

(1) inhibition, inhibitor

(2) competition, substrate

(3) initiation, promoter

(4) None of the above

15) Statement **correct** with reference to enzymes-

(1) Apoenzyme = Holoenzyme + Coenzyme

(2) Holoenzyme = Coenzyme + Cofactor

(3) Holoenzyme = Apoenzyme + Coenzyme

(4) Coenzyme = Apoenzyme + Holoenzyme

16) **Statement-1:** Co-factor play special role in the catalytic activity of enzyme.
Statement-2: Catalytic activity is lost when co-factor is removed

(1) Both statements are correct

(2) Statement 1 is correct and 2 is incorrect

(3) Statement 1 is incorrect and 2 is correct

(4) Both statements are incorrect

17) An antibiotic resistance gene of plasmid vector which get inactivated due to insertion of alien DNA, helps in the selection of-

(1) Transformants
(2) Recombinants
(3) Non-Transformants
(4) 2 & 3 both

18)

In gel electrophorosis, at which end of the gel the sample is loaded?

(1) In the wells
(2) Towards positive electrode
(3) Towards negative electrode
(4) 1 & 3 both

19)

A **selectable marker** is used to :

(1) Help in eliminating the non-transformants and facilitates growth of transformant.
(2) Identify the gene for a desired trait in an alien organism.
(3) Select a suitable vector for transformation in a specific crop.
(4) Mark a gene on a chromosome for isolation using restriction enzyme.

20) A. In elution the separated bands of DNA are cut out from agarose gel and extracted from the gel piece.
B. Cloning vector pBR 322 shows several recognition sites and antibiotic resistance genes.
C. Competent bacterial cell can take up the recombinant DNA.

(1) All are correct
(2) All are incorrect
(3) Except C all are correct
(4) Only A is correct

21) Which of the following steps is/are catalysed by *Taq* polymerase in a PCR?

(1) Denaturation of template DNA
(2) Extension of primer end on template DNA
(3) Annealing of primers to template DNA
(4) Renaturation of template DNA

22) Identify the correct statement w.r.t. normal *E.coli* cells?

(1) They are having genes encoding for resistance to ampicillin
(2) They are having genes encoding for resistance to tetracycline
(3) They are having genes encoding for resistance to chloramphenicol
(4) They do not carry any resistance against any of these antibiotics

23) The first discovered restriction endonuclease which was isolated and characterised 5 years later is

(1) *Hind* III

(2) *Eco*R I

(3) *Hind* I

(4) *Hind* II

24) How many system has to be present on ideal bioreactor
i. Agitator system
ii. An oxygen delivery system
iii. Foam control system
iv. Ph control
v. sampling port system

(1) i, ii, iii only

(2) ii, iii, vi only

(3) iii, iv, v

(4) i, ii, iii, iv, v

25) Which of the following steps is the correct sequence for Recombinant DNA technology?

(1) Isolation of DNA, ligation into a vector, fragmentation by restriction endonucleases, transferring into the host

(2) Fragmentation of DNA by restriction endonucleases, isolation of DNA, ligation of the DNA fragment into a vector, culturing the host cells

(3) Isolation of DNA, fragmentation of DNA by restriction endonucleases, ligation of the DNA fragment into a vector, transferring the recombinant DNA into the host

(4) Ligation of the DNA fragment into a vector, isolation of DNA, transferring the recombinant DNA into the host, culturing the host cells

26) Which enzyme is crucial for linking an antibiotic resistance gene with a plasmid vector to create recombinant DNA?

(1) DNA polymerase

(2) Restriction endonuclease

(3) DNA ligase

(4) RNA primase

27) Assertion A: Blue coloured colonies in the presence of a chromogenic substrate indicate that the plasmid in the bacteria has an insert. Reason R: Insertional inactivation of the β-galactosidase gene due to the insertion of recombinant DNA results in the inability to produce the enzyme necessary for the colour reaction.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

28) If gene of interest was inserted at PvuI site in pBR322 the resulting plasmid will confer resistance to :

(1) Ampicillin
(2) Tetracycline
(3) Kanamycin
(4) Both (1) and (3)

29) A protein encoding gene expressed in a heterologous host will produce a protein product which will be called

(1) Native protein
(2) Recombinant protein
(3) Non-recombinant protein
(4) Restriction enzyme

30) Which of the following is used as a best genetic vector in plants ?

(1) *Pseudomonas putida*
(2) *Bacillus thuringeinsis*
(3) *Rhizobium*
(4) *Agrobacterium tumifaciens*

31) Which of the following statements is false?

(1) Dense irregular connective tissue that has fibroblasts and mostly fibres that are oriented differently are present in skin
(2) Compound epithelium covers the dry surface of the skin and has protective function
(3) Epithelial tissue is rich in intercellular matrix and is vascular
(4) Skin is connected to muscles by areolar connective tissue

32) Which is responsible for controlling the copy number of linked DNA in a plasmid ?

(1) Ori
(2) Selectable marker
(3) Cloning site
(4) Restriction endonuclease

33) Introduction of recombinant DNA into the host cell is a cumbersome process. Although new developments are being introduced to make genetic engineering easier and more accessible. One of the processes by which DNA coated micro-particles are bombarded with high velocity on plant cells is known as

(1) Micro-injection
(2) Biolistics
(3) Transformation
(4) Spooling

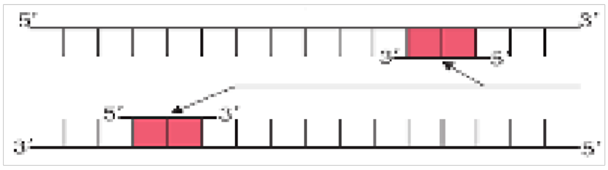
34) The scientists credited with the construction of the first recombinant DNA are

(1) Norman Borlaug and M. S. Swaminathan
(2) Banting and Best
(3) Stanley Cohen and Herbert Boyer
(4) James Watson and Francis Crick

35) The first letter of the name of restriction endonuclease came from the :-

(1) genus of organism
(2) species of organism
(3) family of organism
(4) class of organism

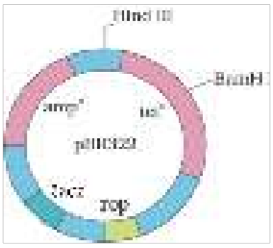
SECTION-2

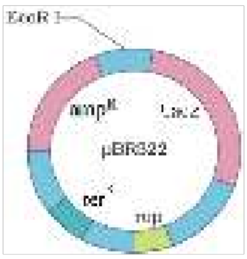
1) 

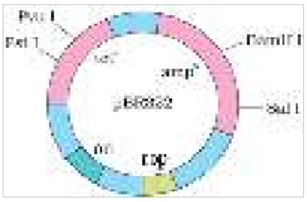
Above diagram showing one stage of PCR. Choose the correct representation.

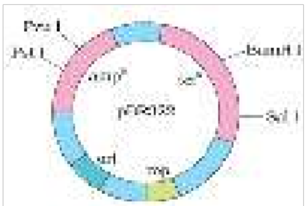
(1) Denaturation
(2) Extension
(3) Binding of DNA primer with DNA template
(4) Polymerisation of deoxyribonucleotides by taq polymerase

2) Which of the following artificial cloning vector of *E.coli* showing ***CORRECT*** sequence of ori site, selectable marker and cloning sites etc.

(1) 

(2) 

(3) 

(4) 

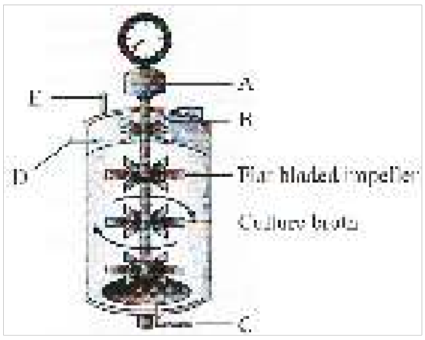
3)

Two enzymes responsible for restricting the growth of bacteriophage in E.coli were isolated in 1963, one of these cut DNA, while other :

(1) Add propyl group to DNA
(2) Add ethyl group to DNA
(3) Add methyl group to DNA
(4) None of the above

4)

Simple stirred-tank bioreactor is given below. Identify A, B, C, D and E :-

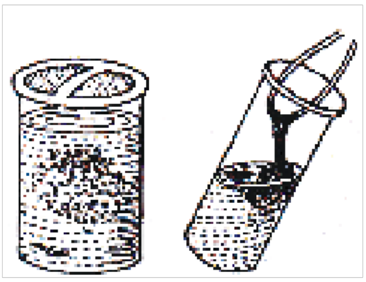


	A	B	C	D	E
(1)	Motor	Foam braker	Sterile air	Steam for sterilization	Acid/Base for pH control
(2)	Foam braker	Sterile air	Steam for sterilization	Acid/Base for pH control	Motor
(3)	Acid/Base for pH control	Motor	Foam braker	Sterile air	Steam for sterilization
(4)	Sterile air	Steam for sterilization	Foam braker	Motor	Acid/Base for pH control

(1) 1
(2) 2
(3) 3
(4) 4

5)

Purified DNA ultimately precipitates out after the addition of chilled ethanol. This can be collected as fine threads in the suspension as seen in the figure. It refers to:



(1) DNA Spooling
(2) DNA digestion
(3) DNA recognition
(4) DNA elution

6)

Which one is associated with downstream processing?
A. Separation B. Purification
C. DNA isolation D. Gene expression

(1) A, B
(2) B, C
(3) C, D
(4) A, B, C, D

7) Select the divalent cation which increases the efficiency with which DNA enters the bacterium through pores in its cell wall.

(1) PO_4^{2-}
(2) Zn^{2+}
(3) Ca^{2+}
(4) Fe^{2+}

8)

DNA is a __1__, molecule and it __2__ pass the plasma membrane :

(1) 1-Hydrophobic, 2-Can
(2) 1-Hydrophilic, 2-Can
(3) 1-Hydrophilic, 2-Cannot
(4) 1-Hydrophobic, 2-Cannot

9) Which of the following are palindromic sequence:

(1) 5' – GAATTC – 3'
3' – CTTAAG – 5'

(2) 5' – TATATA – 3'
3' – ATATAT – 5'

(3) 5' – GAGCTC – 3'
3' – CTCGAG – 5'

(4) All of these

10) Fill in the blanks with correct words
Protein is ___A___ and Lysine amino acid is ___B___ amino acid.

(1) Homopolymer, Basic
(2) Heteropolymer, Acidic
(3) Heteropolymer, Neutral
(4) Heteropolymer, Basic

11) Find out the incorrect match :

	Component	**% of the total cellular mass**
(1)	Water	70 – 90
(2)	Protein	3
(3)	Lipid	2
(4)	Nucleic acid	5 – 7

(1) 1
(2) 2
(3) 3
(4) 4

12) In all the connective tissues except one, the cells secrete fibres of structural proteins called collagen or elastin, the tissue is

(1) Bone
(2) Cartilage
(3) Blood
(4) Adipose tissue

13) Areolar tissues contain

(1) T lymphocytes tissue and lymphocytes
(2) Fibroblast, macrophages, mast cells
(3) Fibroblast cells only
(4) Fibroblasts and fat globules

14) Cartilage is present in :-

(1) Between adjacent bones of vertebral column and limbs
(2) In middle of the long bone
(3) Both (1) and (2)
(4) None of these

15) Match the columns-I and II, and choose the correct combination from the options given :-

Column-I		Column-II	
a.	Adhering Junctions	I.	Help to stop substances from leaking across a tissue
b.	Gap Junctions	II.	Perform cementing to keep neighbouring cells together
c.	Tight Junctions	III.	Facilitate the cells to communicate with each other

(1) a-III, b-II, c-I

(2) a-II, b-III, c-I

(3) a-II, b-I, c-III

(4) a-I, b-III, c-II

ANSWER KEYS

PHYSICS

SECTION-1

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**4**	**1**	**1**	**1**	**2**	**4**	**3**	**1**	**2**	**4**	**1**	**3**	**1**	**1**	**2**	**3**	**1**	**2**	**3**	**3**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
A.	**4**	**1**	**4**	**1**	**1**	**4**	**1**	**1**	**3**	**2**	**2**	**2**	**2**	**1**	**3**					

SECTION-2

Q.	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50
A.	**2**	**2**	**3**	**1**	**2**	**1**	**2**	**4**	**1**	**2**	**3**	**2**	**2**	**3**	**4**

CHEMISTRY

SECTION-1

Q.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
A.	**3**	**4**	**2**	**2**	**1**	**4**	**3**	**4**	**3**	**3**	**4**	**1**	**3**	**4**	**4**	**1**	**4**	**3**	**2**	**3**
Q.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85					
A.	**4**	**3**	**2**	**3**	**4**	**2**	**3**	**3**	**2**	**4**	**2**	**2**	**2**	**3**	**1**					

SECTION-2

Q.	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
A.	**2**	**1**	**3**	**1**	**1**	**2**	**2**	**2**	**3**	**1**	**3**	**1**	**1**	**1**	**1**

BOTANY

SECTION-1

Q.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	**3**	**2**	**1**	**4**	**3**	**3**	**4**	**4**	**3**	**3**	**2**	**2**	**2**	**1**	**2**	**1**	**4**	**1**	**3**	**1**
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135					
A.	**3**	**3**	**4**	**1**	**1**	**1**	**4**	**2**	**3**	**4**	**4**	**1**	**3**	**2**	**3**					

SECTION-2

Q.	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**1**	**3**	**4**	**1**	**4**	**1**	**2**	**2**	**2**	**1**	**1**	**3**	**3**	**4**	**1**

ZOOLOGY

SECTION-1

Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	1	3	4	4	1	3	1	1	2	2	2	1	3	1	2	4	1	1

Q.	171	172	173	174	175	176	177	178	179	180	181	182	183	184	185
A.	2	4	4	4	3	3	4	2	2	4	3	1	2	3	1

SECTION-2

Q.	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200
A.	3	4	3	1	1	1	3	3	4	4	2	3	2	1	2

SOLUTIONS

PHYSICS

1) Dimension of

$$X = \frac{[M^{-1}L^{-3}T^{4}A^{2}]\,[L^{1}]\,[M^{1}L^{2}T^{-3}A^{-1}]}{T^{1}}$$

Dimension of X $= \dfrac{T^{4}\,T^{-3}\,A^{1}}{T^{1}}$

$= [A^{1}]$ = Current

4) Slope of v-t graph ⇒ acceleration

Maximum slope = Maximum acceleration

7) $F = 6\pi\,\eta r v$

hence it is dimensionally correct but numerically incorrect

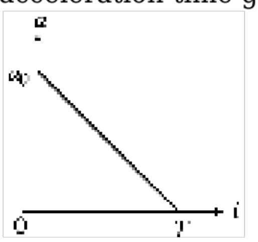
9) To catch $(v_y)_A = (v_y)_B$

$\Rightarrow v\sin 37° = 30$

$\Rightarrow v = \dfrac{30}{3/5} = 50\text{m/s}$

Time taken $\dfrac{200}{v\cos 37°} = 5\text{s}$

11) Given acceleration-time graph is shown in figure.



Let $a(t = 0) = a_0$ = constant

From the graph, we have

$$\frac{t}{T} + \frac{a}{a_0} = 1$$

$\Rightarrow \quad a_0 t + aT = a_0 T \qquad ....(i)$

Also, we know that,

$a = \dfrac{dv}{dt} \qquad ....(ii)$

From Eqs. (i) and (ii), we get

$$\frac{a_0 T - a_0 t}{T} = \frac{dv}{dt}$$

Integrating both sides, we get

$$\int_0^t (a_0 T - a_0 t)\,dt = T\int_0^v dv$$

$\Rightarrow \quad a_0Tt - \frac{a_0t^2}{2} = Tv \Rightarrow v = a_0t - \frac{a_0}{2T}t^2$

Therefore, v-t graph must be parabolic.

Now, at t = 0, v = 0

$$t = T, v = \frac{a_0T}{2} = \text{constant}$$

These conditions are satisfied by graph (a).

15)

$R = \frac{2U_xU_y}{g} = \frac{2 \times 6 \times 8}{10} = 9.6$

19) $\vec{V}_r = 2\hat{i} - 3\hat{j}$

$\vec{V}_m = V\hat{i}$

$\vec{V}_{rm} = (2 - V)\hat{i} - 3\hat{j}$

$(2-V)^2 + 3^2 = 5^2$

$(2-V)^2 = 16$

$2 - V = \pm 4$

$V = -2$ and $V = 6$

So, $\vec{V}_m = -2\hat{i}$ and $\vec{V}_m = 6\hat{i}$

20)

N = x cm

b = n

a = n - 1

L.C. $= M\left(\frac{b-a}{b}\right)$

$= x\left(\frac{n-(n-1)}{n}\right)$

$x\left(\frac{n-n+1}{n}\right)$

$\text{L.C.} = \frac{x}{n}$

21) Least count $= \frac{\text{Pitch}}{\text{No. of division of C.S.}}$

L.C. $= \frac{1\text{mm}}{100} = 0.01 \text{ mm} = 0.001 \text{ cm}$

Diameter of the wire = L.S.R + (C.S.R × L.C.)

= 0 + (52 × 0.001)

= 0.052 cm

23) $\because \lambda = \frac{h}{P}$, nucleus is break in two parts the momentum is conserved then $\lambda_1 : \lambda_2 = 1 : 1$

Telegram: @NEETxNOAH

26) $\frac{\Delta x}{x} = 1\% = 10^{-2}$

$\Rightarrow \frac{\Delta y}{y} = 3\% = 3 \times 10^{-2}$

$\frac{\Delta z}{z} = 2\% = 2 \times 10^{-2}$ $\Rightarrow t = \frac{xy^2}{z^3}$

$\frac{\Delta t}{t} = \frac{\Delta x}{x} + \frac{2\Delta y}{y} + \frac{3\Delta z}{z}$

$= 10^{-2} + 2 \times 3 \times 10^{-2} + 3 \times 2 \times 10^{-2}$

$= 13 \times 10^{-2}$

 % error in t $= \frac{\Delta t}{t} \times 100 = 13\%$

31) $\frac{dx}{dt} \Rightarrow$ velocity

$m\frac{d^2x}{dt^2} = ma \Rightarrow$ force

$\int Vdt \Rightarrow$ displacement

$\int a\,dt \Rightarrow$ velocity

32) 20 VSD = 19 MSD

$1\text{ VSD} = \frac{19}{20}\text{MSD}$

L.C. = 1 MSD - 1 VSD

$0.2\text{ mm} = 1\text{ MSD} - \frac{19}{20}\text{MSD}$

0.2 × 20 mm = 1 MSD

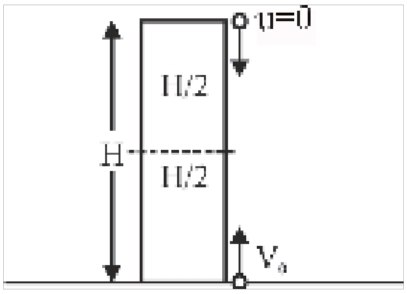
$\boxed{\text{MSD} = 4\text{ mm}}$

34) $r_0 = \frac{ze^2}{4\pi\varepsilon_0\left(\frac{1}{2}m_\alpha v^2\right)}$

35) $\vec{a}_c = \frac{v^2}{r}\left(-\cos\theta\,\hat{i} - \sin\theta\hat{j}\right)$

$= \frac{2^2}{4}\left(-\cos 45\hat{i} - \sin 45\hat{j}\right)$

$= -\left(\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

36) 

Time of meeting $t = \frac{H}{V_0 + 0}$

$t = \frac{H}{V_0}$ ...(1)

For ball from ground using eq. (2)

$S = ut + \frac{1}{2}at^2$

$\frac{H}{2} = V_0(t) - \frac{1}{2}g(t^2)$ ...(2)

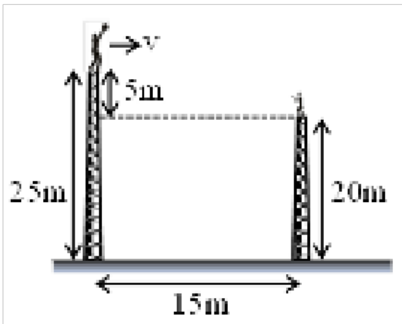
$\frac{H}{2} = V_0\left(\frac{H}{V_0}\right) - \frac{1}{2}g\left(\frac{H}{V_0}\right)^2$

$V_0^2 = gH$ $V_0 = \sqrt{gH}$

37)

For vertical disp.

$5 = 0 + \frac{1}{2} 10t^2$



t = 1 sec

and,

R = V × t

15 = V × 1

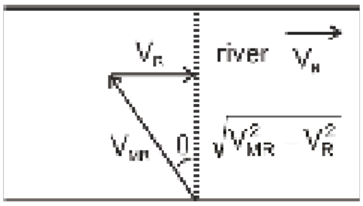
V = 15 m/s

38) $E_1 = \frac{13.6eV}{n^2}$

n = 2

E2 = 3.4 eV

40) 15 min = 1/4 hr.



$t = \frac{d}{V_y}$

$\Rightarrow \frac{1}{4} = \frac{1}{\sqrt{V_{MR}^2 - V_R^2}} = \frac{1}{4} = \frac{1}{\sqrt{5^2 - V_R^2}}$

$\Rightarrow V_R = 3$ km/h

41) $H = \frac{v^2 \sin^2\theta}{2g}$ and $R = \frac{v^2 \sin 2\theta}{g}$

Since, R = 2H, so $\frac{v^2 \sin 2\theta}{g} = 2 \times \frac{v^2 \sin^2\theta}{2g}$

or $2 \sin\theta \cos\theta = \sin^2\theta$ or $\tan\theta = 2$

 $R = v^2 \times \frac{2}{g} \times \sin\theta \cos\theta$

$= \frac{2v^2}{g} \times \frac{2}{\sqrt{5}} \times \frac{1}{\sqrt{5}} = \frac{4v^2}{5g}$.

42) $\frac{hC}{\lambda_3} = \frac{hC}{\lambda_1} + \frac{hC}{\lambda_2} \Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$

43) $\tan\theta = \frac{v_y}{v_x} = \frac{gt}{v_x}$

$\tan 30^\circ = \frac{10 \times t}{40}$

 $t = \frac{4}{\sqrt{3}}$ sec

44) In given equation, $\frac{\alpha z}{k\theta}$ should be dimensionless

 $\alpha = \frac{k\theta}{z} \Rightarrow [\alpha] = \frac{[ML^2T^{-2}K^{-1} \times K]}{[L]} = [MLT^{-2}]$ and

$P = \frac{\alpha}{\beta} \Rightarrow [\beta] = \left[\frac{\alpha}{p}\right] = \frac{[MLT^{-2}]}{[ML^{-1}T^{-2}]} = [M^0L^2T^0]$

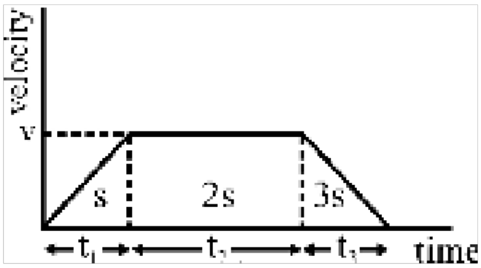
45) $T = \frac{2(15)\sin 37^o}{g} = 1.8$ sec

speed $V = \frac{9}{1.8} = 5$ sec

47)

As $t = x^2 + x \Rightarrow dt/dx = 2x + 1$
$\Rightarrow \quad v = (2x + 1)^{-1} \qquad \ldots$ (i)
[as dx/dt = v], Now by chain rule
$a = \frac{dv}{dt} = \frac{dv}{dx}\cdot\frac{dx}{dt} = v\frac{dv}{dx}$
$\Rightarrow \quad a = (2x + 1)^{-1}\frac{d}{dx}(2x + 1)^{-1}$
$= -2\ (2x + 1)^{-3}$
So, Retardation $= -a = 2\ (2x + 1)^{-3}$

48) $v_{avg} = \frac{6s}{t_1 + t_2 + t_3}$
$v_{avg} = \frac{6s}{\frac{2s}{v} + \frac{2s}{v} + \frac{6s}{v}}$
$v_{avg} = \frac{6s \times v}{10s} \Rightarrow \frac{v_{avg}}{v} = \frac{3}{5}$



50) NCERT Reference: Class XI, Part I, Page No. 76

CHEMISTRY

51) Limiting reagent A
mole of C $= \frac{1}{3}$ mol
mass of C $= \frac{1}{3} \times 24 = 8$ g

52) Weight of NH_3 = 4.25 g
Number of mole of
$NH_3 = \frac{\text{Weight}}{\text{Molecular weight}}$
$= \frac{4.25}{17} = 0.25$ mol
Number of molecules in 0.25 mole of NH_3
$= 0.25 \times 6.023 \times 10^{23}$
So, number of atoms

$= 4 \times 0.25 \times 6.023 \times 10^{23} = 6.0 \times 10^{23}$

55) According to the given reaction,
Number of moles of $CaCO_3$,

$= \frac{1}{2} \times$ mole of HCl

$= \frac{1}{2} \times 0.5 \times \frac{50}{100} = 0.0125$ mol

Mass of pure $CaCO_3 = 0.0125 \times 100$
= 1.25 g

Mass of impure sample

$= \frac{1.25 \times 100}{95} = 1.32$ g

58) Electron doesn't exist in nucleus.

59) $\Delta E \propto \frac{1}{\lambda}$; As n↑, ΔE↓
On absorption of photon e^- gets excited to higher energy levels.

64)

Last electron goes in (n-2) f-subshell

65) Due to lanthanide contraction the size of atoms in options 1, 2 & 3 are approxdimately same.

67)

H — F
EN ⇒ 2.1 4.0 ΔEN = 1.9 most polar

71) Aldehyde (aliphatic and aromatic) show reaction with Tollen's reagent and aliphatic aldehydes show Fehling solution, but Ketones do not show there reactions.

72)

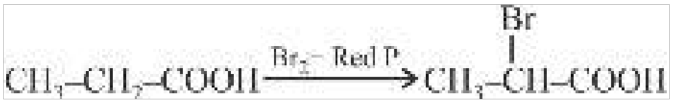
$CH_3CH_2CH_2COOH > CH_3CH_2CH_2CH_2OH > CH_3CH_2CH_2CHO$

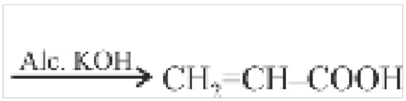
(III)	(I)	(II)
⇓	⇓	⇓
Dimer formation due to intermolecular H-bonding	Intermolecular H-bonding	H-bonding is absent

74)

CrO_2Cl_2

76) 

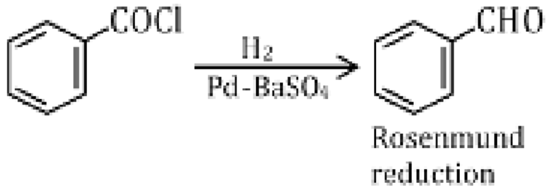
77) 



78) Primary and secondary amine differentiated by all three test.

79)

Hoffmann bromamide degradation reaction.



85)

86) $\frac{w}{M_W} = \frac{V}{22.4}$

$\frac{60}{M_W} = \frac{5.6}{22.4} \Rightarrow M_W = 240$

$V.D. = \frac{M_W}{2} = 120$

91)

Electron configuration of M^{2+}

$= (Ar)\ 3d^{10}\ 4s^2$

no. of electron in m = 18 + 10 + 2 + 2 = 32 (Ge)

98)

Fact (NCERT-Nitrogen compound)

BOTANY

101) Module No.6 Pg.# 189

102) NCERT 10.2.3 PG. 165-166

103) NCERT PG. 168-169

105) NCERT–XI Pg. # 168

106) NCERT (XI) Pg. # 139

108) NCERT Pg. # 165

109) NCERT Pg # 168

112) NCERT Pg. # 166

126) NCERT Page # 88

132)

NCERT Pg. # 136

133) NCERT XI Pg # 137, 138

134) NCERT Pg. # 136, 8.5.7

136) NCERT PG. 168

138) NCERT (XIth) Pg. # 166

139) Module No-6, P. No # 222

146)

NCERT Pg # 126

147) NCERT Pg#132

148) NCERT XI Pg # 132, 133

150) NCERT Pg#138

ZOOLOGY

154)

NCERT XII. Pg. No. 107

155)

Ncert Page 146.

156) NCERT (XI) Pg. # 145

157) NCERT-XI Pg. # 148 para 2

158) The assertion is true: amino acids can exist in a zwitterionic form, which is a dipolar ion with both positive and negative charges due to the ionization of amino and carboxyl groups. However, the reason is false because the structure of amino acids indeed changes in solutions of different pH, as they can gain or lose protons, leading to different ionic forms. Therefore, A is true, but R is false.

159) Statement I: Incorrect. While paper made from plant pulp and cotton fibre is indeed cellulosic, it is not necessarily composed of complex polysaccharides; rather, it is primarily composed of cellulose, which is a simple polysaccharide.
Statement II: Correct. Chitin is a complex polysaccharide that is mostly a homopolymer, and it is found in the exoskeletons of arthropods.

160) NCERT-XI, English Page # 157, (Last Para) and Fig.9.7
हिन्दी पेज # 157, (नीचे से दूसरा पेराग्राफ) एवं पेज # 158 (चित्र 9.7)

163) NCERT EDITION 2022 - 2023 Pg. No. 158

172) NCERT P. No. 199

173) *Hind* II was the first RE which was isolated and characterised 5 years later.

175) The correct sequence of steps in Recombinant DNA technology starts with the isolation of DNA to ensure it is free from other macromolecules. This is followed by fragmentation of the DNA using restriction endonucleases, after which the desired DNA fragment is ligated into a vector. The recombinant DNA is then transferred into the host for further processing.

176) DNA ligase is the enzyme that acts on cut DNA molecules and joins their ends. This is essential in the process of forming recombinant DNA, where an antibiotic resistance gene is linked with a plasmid vector to create a new combination of circular autonomously replicating DNA in vitro.

177)

The assertion is false because blue coloured colonies indicate that the plasmid in the bacteria does not have an insert, as the β-galactosidase gene is active and can metabolize the chromogenic substrate to produce the blue colour. The reason is true as insertional inactivation of the β-galactosidase gene due to the insertion of recombinant DNA does result in the inability to produce the enzyme, but this would lead to white colonies, not blue. Therefore, A is false, but R is true.

178) NCERT Page # 199

182) NCERT Pg. # 199

186) NCERT-XII, Pg. # 202

187)

NCERT-XII, Page No. 199

193)

NCERT, Pg # 200

195) NCERT Pg # 147

197) NCERT Pg.# 103

198) NCERT Page: 103